

PAPER_1321 — Origin of Stellar Magnetic Fields

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "stellar_magnetism_origin"})`

Master Expression

B emerges from SCm phonon $\times$ dynamo amplification via K_{MEX}

Match: mechanism

Interpretation

Sun $\sim 1\text{G}$, magnetar $\sim 10^{15}\text{G}$ — both via SCm condensate $\times$ $\text{OMEGA_SCM} = 1.25\text{ THz}$ dynamo coupling. $F_{\text{U_Bi_i}}$ buoyancy provides shear.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $\text{SO}_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 11, 2026, Youngstown, OH.